DOCTOR OF PHILOSOPHY

IN

COMPUTER SCIENCE AND APPLICATIONS

OFFERED BY

PANJAB UNIVERSITY, CHANDIGARH
(Examination 2017 - 2018)

Ph.D. Programme

Ph.D. Programme

The purpose of the Doctor of Philosophy programme in Computer Science and Applications is to prepare students for research and teaching positions in universities and colleges, and for research and leadership positions in industry and government. The primary aim of the programme is to train students in the methods of scientific inquiry and independent research. This is accomplished through advanced coursework and active participation with the faculty in their research programmes. Doctor of Philosophy students are expected to have a broad knowledge of all fields of computer science and have a deep understanding of at least one of its areas. In addition to this requirement, a Doctor of Philosophy student must be up to date in all the developments in his/her major area of specialization. The most important component of the Doctor of Philosophy programme is learning to perform independent and significant research in one's area of specialization.

Scheme and Syllabus

First Semester

Sr. No.	Course No.	Course Title	L	T	P	Credits
1	CSA - 101	*Research Methodology*	**3**	**-**	**1**	**3**
2	CSA - 102	*Research Seminar*	**2**	**-**	**-**	**2**
3	**Elective I:** CSA -103 *Advanced Software Engineering* CSA -104 *Soft Computing Techniques* CSA -105 *Multiagent Decision Making Systems* CSA- 106 *Image Procession and Computer Vision* CSA -107 *Educational Technologies and Advanced Database Systems* *CSA-108 Mobile Communication and Android Application Development* *CSA-109 Parallel and Distributed Computing* *CSA-110 Data Mining and Data Warehousing* *CSA-111 Simulation & Modeling* *CSA-112 Network Technologies*		**3**	**-**	**1**	**3**

Research Methodology

Paper Code: **CSA 101**

Objectives: The main objective of the course is to apprise the researchers of the research methodologies, tools and techniques for effective research.

SECTION – A

Introduction: Objectives of Research, Types of Research, Steps in Research, characteristics and Criteria of good research, problems versus research problems, sources of research problems

Formulating the Research problem: reading research papers, Literature Review based on national and international journals, theses, reports etc, writing about the literature reviewed with critical analysis, identifying gaps (if any) in the existing research based on the reviewed literature, considerations in selecting a research problem, steps in formulating a research problem, formulating a research problem in applied, qualitative and quantitative research, proposed research goals and objectives

Research design: Need for research design, features of a good design, different research designs, developing a research plan, designs commonly used in qualitative research, designs used in quantitative research, research design for application specific research, sample design and types, random samples; case studies of qualitative, quantitative and applied research problems and their design

Data Collection and Methods: Primary and secondary data sources, techniques for collection of data using primary and secondary sources, methods of data collection in qualitative, quantitative and applied research, standard databases, self created databases, methods of validity, reliability and verification of output or research findings.

SECTION – B

Processing and Analysis of Data: Role of statistics in research, fundamentals of statistical analysis and inference, correlation and Regression, probability distributions, hypothesis testing, parametric and non parametric tests, Principal Component Analysis, Moments and response curve methods, state vector machines, uncertainty analysis and Bayesian networks; Generalization and interpretation.

Using data analysis tools: Understanding the features and applications of various tools including SPSS, R and MATLAB; using these tools for processing and analysis of data using various techniques mentioned above; tabular and graphical description of data using these tools.

Research Report Writing: Significance of reports, types of reports, steps in writing a report, structure and components of scientific reports, Importance of effective communication, Bibliography and Referencing, standard formats for referencing including IEEE and Springer, Oral presentation and role of visual aids, basic presentation skills, writing a research paper; use of documentation and presentations tools such as LATEX, SLITHY and Microsoft Office.

Ethics in research: Principles for research ethics, legal issues, ethical issues, copyright, Intellectual property rights and patents law, plagiarism, citation and acknowledgement.

Developing a research proposal: Minor and major research projects, scope of the project, applications and advantages, goal and objectives of proposed project, proposed model, design, and implementation tools, testing and validation, financial requirements.

Reference Books:

1.	C. R. Kothari	:	Research Methodology – Methods and techniques (Second Revised Edition) New Age International Publishers 2009
2.	Ranjit Kumar	:	Research Methodology: A step by step guide for Beginners, Sage Publications Ltd., 2014
3.	Wayne Goddard and Stuart Melville	:	Research methodology: An Introduction for science & Engineering students, Juta and Company Ltd., 2004
4.	Stuart Melville and Wayne Goddard	:	Research methodology: An Introduction, Juta and Company 1996.
5.	Sreenivasulu N. S.	:	Law relating Intellectual Property, Partridge Publications, 2013
6.	R. pannerselvam	:	Research Methodology, PHI, 2005
7.	Paul Connolly	:	Quantitative Data Analysis in Education: A Critical Introduction using SPSS
8.	Kiran Singh, B.B. Chaudhuri	:	MATLAB Programming, PHI
9.	James Gips	:	Mastering MS Excel 2007: A Problem Solving Approach
10.	Amos Gilat	:	MATLAB: An Introduction with Applications, 3rd Edition, 2008, Wiley.
11.	Leslie Lamport		Latex: A document preparation system: User's guide and Reference Manaual, Second Edition, 2007

Research Seminar

Paper Code: **CSA 102**

Objectives : This course is a seminar course for Research Scholars in Computer Science and Applications. The aim is to explore literature including recent papers of interest from reputed and refereed Journals, theses and other relevant documents and understand the state of the art research in Computer Science. Each student should identify a base paper related to the proposed research area after making extensive literature review, prepare a report on the paper, submit the document and present the same in the research seminar.

SECTION - A

Understanding concepts in the proposed area of research: Identify one or two books that cover the proposed area to be enriched with sufficient background to follow the research papers

Literature survey: Significance of literature survey; using online tools such as Google, CiteSeer, ACM digital library, IEEE, The online computer science bibliography; searching survey papers and other relevant and standard material in the proposed area of research

How to study a research paper: Reading abstract to find what is covered in the paper, getting more background from study of papers given in references, understanding design methodology and techniques used and other techniques that could have been used, implementation tools, testing criteria and performance analysis.

SECTION - B

Preparing Report on the research paper identified for the seminar: Reasons for selecting the paper after extensive literature review, keywords and background required to understand the paper, abstract, literature review, design techniques, implementation and results, conclusions and scope for further work and references in own words of the candidate with detailed explanation as required. The advantages and drawbacks of the paper are also to be included in the report. The report reflects the candidate's analytical and report writing skills.

Presentation: The presentation must include the contents of the report. The contents of the report must be explained briefly but including the required technical details. The presentation reflects the candidate's communication skills, explanation capacity and question answering capability

Suggested Reading

- library.concordia.ca/help/writing/literature-review.php
- www2.le.ac.uk
- www.eecs.harvard.edu/~michaelm/postscripts/ReadPaper.pdf
- www.bmj.com/about-bmj/resources-readers/publications/how-read-paper
- http://www.wikihow.com/Write-a-Research-Paper
- www.wikihow.com/ **Write-a-Report**
- https://www.nfer.ac.uk/schools/research-in-schools/presenting-your-findings/
- www.wikihow.com/**Give-a-Seminar**

Advanced Software Engineering

Paper Code: **CSA 103**

Objectives: This comprehensive course provides a broad perspective on the advanced topics of Software Engineering and embark on more in-depth research or practice in software engineering.

SECTION – A

Software Architecture - Concepts, representation techniques, development methods, and tools for architecture-centric software engineering, decentralized software architecture, architectures for Mobile and Pervasive Systems evaluating software architectures,

Software Configuration Management: Configuration Management, Change Management, Version Management, Build and Release Management.

Software Reuse: Motivation, Inhibitors, Techniques - Component Based Software Engineering, Process Models, Reuse Metrics.

Software Re-engineering: Introduction, Restructuring and Reverse Engineering, Reengineering existing systems, Data Re-engineering and migration, Reengineering Metrics.

SECTION – B

Software Project Management: Software Project Planning, Conventional Software Management, Evolution of Software Economics, Improvement of Software Economics.

Quality: Quality issues in the life-cycle model. Quality planning and management. Aspects of quality: reliability, maintainability, correctness, usability. Risk analysis and management.

Measurement: The fundamentals of Measurement Theory. Introduction to Metrics (Quality metrics, cost metrics and process metrics). Function points and object points. Estimation Models and Techniques (Top-down/Bottom Up, Analytical Models (Cocomo, Putnam, etc.), Custom Models.

Testing: Requirements, Unit, integration and system testing, validation. Structural testing: coverage techniques. Behavioral testing: domain testing, finite state testing. Mutation and fault seeding. Tools and instrumentation.

Books:

1. Mary Shaw and David Garlan. : Software Architecture: Perspective on an Emerging Discipline, Prentice –Hall.
2. Paul Clements, Rick Kazman, and Mark Klein. : Evaluating Software Architecture Methods and Case Studies. Addison Wesley.
3. Len Base, Paul Clements, Rick Kazman. : Software Architecture in Practice. Second Edition, Addison Wesley.
4. I. Sommerville : Software Engineering, Addison Wesley.
5. Walker Royce Software project management, , Pearson Education.
6. Stephen T. Albin : The Art of Software Architecture: Design methods and Techniques, Wiley
7. David Budgen : Software Design, 2nd edition, Addison-

Wesley
8. William A. Florac and Anita D. Carleton — Measuring the Software Process; Addison-Wesley
9. Fenton, Norman E., Whitty & Iizuka — Software Quality Assurance & Measurement: A Worldwide Perspective, Thomson Computer Press
10. Horch J. W — Practical Guide to Software Quality Management, Artech House Computer Science
11. Kan Stephen H. — Metrics and Models in Software Quality Engineering, Addison-Wesley.
12. Glenford J. Myers — The Art of Software Testing, John Wiley & Sons
13. Robert S. Arnold — Software Re-engineering, Robert S. Arnold, IEEE Comp. Society Press.

Soft Computing Techniques

Paper Code: **CSA 104**

Objectives: The objective of the course is to provide understanding of the soft computing techniques including artificial neural networks, fuzzy logic and genetic algorithms useful for research in intelligent systems and image processing. After going through the course, the candidate will be able to choose and apply the techniques required for the research problem undertaken.

SECTION - A

Introduction to Soft Computing: Basic soft computing techniques: Neural networks, Fuzzy logic, Genetic algorithms; Hybrid systems, Application of soft computing.

Artificial Neural Networks: Fundamentals, Characteristics of neural networks, Basic models of artificial neural network: Connections, Learning method, Activation functions; Neural network architectures; Applications of neural networks.

Supervised Learning Networks : Introduction to supervised learning, Architecture and training algorithms for perception network and back propagation networks

Unsupervised Learning Networks: Introduction to unsupervised learning, Architecture and training algorithms for Kohonen self-organizing maps and adaptive resonance theory networks (ART1, ART2).

Hopfield Networks: Introduction, Architecture and training algorithm for discrete Hopfield nets.

SECTION - B

Fuzzy Logic: Introduction and Application to Fuzzy logic, Classical sets, Fuzzy Sets, Operations on Fuzzy Sets, Fuzzy Relations, Membership functions, Methods of membership value assignment, and Defuzzification methods, Neuro-fuzzy systems, genetic algorithms and fuzzy logic

Fuzzy Logic Control Systems: Architecture and Operation of Fuzzy Logic Control (FLC) systems, FLC System Models, Applications of FLC Systems.

Introduction to genetic algorithms: Biological background, Genetic algorithms versus traditional algorithms. Basic terminologies in genetic algorithm: genes, fitness and populations. Steps in a general genetic algorithm

Operations in genetic Algorithm: Selection, Crossover and Mutation, Applications of Genetic Algorithms

Soft computing Tools: features, applications, and use of tools such as MATLAB, SCILAB, and WEKA, developing small projects using these tools and employing soft computing techniques

Suggested Readings:

1. S.N. Sivanandam and S.N. Deepa: Principles of Soft Computing, Second Edition, John Wiley.
2. Karray and Silva: Soft Computing and Intelligent Systems Design, Pearson Education.
3. Gallant Stephen: Neural Network Learning and Expert System, MIT Press.

4. S. Rajasekaran and Pai: Neural Networks, Fuzzy Logic, and Genetic Algorithms, PHI.
5. Rao and Rao: C++ Neural Networks & Fuzzy Logic, BPB
6. Timothy Ross, Fuzzy Logic with Engineering Applications, Third Edition,Wily Publications, 2010
7. Russel C. Eberhart, Yuhui Shi, Computational Intelligence: Concepts to Implementation, Morgan Kaufman, 2007

Multiagent Decision Making Systems

Paper Code: **CSA 105**

Objectives: The main objective of the course is to provide understanding of distributed artificial intelligence and multiagent systems, applications, agent communication protocols and tools and techniques for developing multiagent decision making systems. After going through the course the candidate will be capable of undertaking research in developing any multiagent decision making system.

SECTION – A

Agents and Multiagent Systems : Limitation of artificial intelligence (AI), introduction and advantages of distributed AI, agents, multiagent systems (MAS), applications, agent oriented programming, examples of agents and multiagent systems

Agent Communication : Communication levels, message types, KQML (Knowledge Query and Manipulation Language), KIF (Knowledge Interchange Format), ontologies, agent interaction protocol, co-ordination and co-operation protocols, contract nets, and black board systems.

Distributed Model for Decision Support : Introduction, agent architecture for distributed decision support system, Case Studies from various areas such as Traffic Management, resource management, ecommerce and disaster management.

Artificial Neural Networks: Introduction to Artificial Neural Networks, Neural Networks Architectures, Supervised and Unsupervised Learning, Perceptron, Back Propagation Networks and Associative Memory.

SECTION – B

Fuzzy Logic: Introduction, Basic Concepts of Fuzzy Logic, Fuzzy Sets, Fuzzy Relations, Fuzzy Graphs, and Fuzzy Arithmetic, Fuzzy If-Then Rules, Fuzzy Implications and Approximate Reasoning, Fuzzy Logic in Control Engineering, Fuzzy Logic and Artificial Intelligence, Fuzzy Logic in Database Management and Information Systems, Fuzzy Logic in Pattern Recognition, Neuro-Fuzzy Systems, Genetic Algorithms and Fuzzy Logic.

Genetic Algorithms: Fundamentals, Encoding, Fitness function, Reproduction, Inheritance Operators, Crossover, Inversion and Deletion, Mutation Operator, and Bit-wise Operators; applications

Bayesian Networks: Reasoning under uncertainty**,** Bayesian Networks, Inference in Bayesian Networks, Decision Networks, and Applications of Bayesian Networks.

Programming using agent development and decision making tools: Features, applications and programming using JADE (Java Agent Development Environment), Java agents, Java agent framework, MATLAB, SCILAB, WEKA, and GENIE, for developing multiagent decision making systems

Reference Books:

1.	Reza Langari, John Yen	:	Fuzzy Logic: Intelligence, Control, and Information, 1st Edition, Pearson (Chapters 1-8, 11-13, 16, 17)
2.	Laurene V. Fausett	:	Fundamentals of Neural Networks: Architectures, Algorithms And Applications, 1993, Prentice Hall. (Chapters 1-6)
3.	Timothy Ross	:	Fuzzy Logic with Engineering Applications, 3rd Edition 2010, Wiley
4.	Kevin B Korb, Ann Nicholson, Korb B Korb	:	Bayesian Artificial Intelligence, 2004, Chapman & Hall/CRC. (Chapters 1-5)
5.	Russell C. Eberhart, Yuhui Shi	:	Computational Intelligence: Concepts to Implementation, 1st Edition, 2007. Morgan Kaufman
6.	David hand, Heikki Mannila, Padhraic Smyth	:	Principles of Data Mining, PHI
7.	S Rajasekharan, G.A. Vijayalakshmi Pai		Neural Networks, Fuzzy Logic and Genetic Agorithms, PHI
8.	Gerhard Weiss		Multiagent Systems edited by, MIT press 1999.
9.	Amos Gilat		Amos Gilat, MATLAB: An Introduction with Applications, 3rd Edition, 2008, Wiley

Image Processing and Computer Vision

Paper Code: **CSA 106**

Objectives: The course enables the students to learn fundamentals of Image processing and computer vision. The candidate after going through the course will be capable of performing research in image processing and computer vision and can develop required algorithms using Matlab and other image processing and computer vision tools.

SECTION – A

Introduction to Image Processing: Digital Image Processing, Elements of Image Processing Systems—Image Acquisition, Processing, Communication, Display.

Digital Image Fundamentals:
Uniform and Non-uniform Sampling and Quantization, Basic Relationships between pixels—Neighbours of a pixel, Connectivity, Distance Measures, Imaging Geometry—Perspective transformations, Camera Model, Stereo Imaging.

Image Transforms: Introduction to Fourier Transform, Discrete Fourier Transform, Properties of the Two – Dimensional Fourier Transform, The Fast Fourier Transform (FFT), Inverse FFT, Walsh, Hadamard and Discrete Cosine Transforms.

Image Enhancement: Histogram Processing, Image Averaging, Smoothing Filters, Sharpening Filters, Low Pass and High Pass Filtering, Generation of Spatial Masks from frequency Domain Specifications, Colour Image Processing—Colours Fundamentals, Colour Models, Pseudo-Colour image processing.

SECTION – B

Image Restoration: Degradation Model, Circulant and Non-circulant Matrices, Algebraic Approach to Restoration, Inverse Filtering, Wiener Filter, Constrained Least Square Restoration, Geometric Transformations.

Image Compression: Fundamentals, Image Compression models and techniques, Low Compression, Image Compressions standards.

Image Segmentation, Representation and Description: Detection of Discontinuities, Edge Linking and Boundary Detection, Hough Transform, Thresholding, Region Oriented Segmentation; Fundamentals of Image Representation and Description

Pattern Recognition and Computer Vision: Fundamentals, Applications, feature extraction techniques including Principal Component Analysis (PCA), Local Binary patterns (LBP), Euclidean distance; Classification techniques including Decision Trees, Support Vector Machines, and Artificial Neural Networks; Image Interpretation and Content based Retrieval

Using Image Processing and Computer Vision Tools: Standard and open source tools including MATLAB, SCILAB, OpenCV, Programming using PYTHON, implementation of image processing and computer vision algorithms using these tools

References

1. Rafael C. Gonzalez, Richard E.Woods, Steven L. Eddins: Digital Image Processing, MGatesmark Publishing, 2009
2. Jain, Anil K.: Fundamentals of Digital Image Processing, Pearson.
3. Jensen, John R.: Introductory Digital Image Processing, Prentice Hall.
4. Dougherty, Edward R.: Image Processing Digital Techniques.

Educational Technologies and Advanced Database Systems

Paper Code: **CSA 107**

Objectives: ***The main objective of the course is to provide a deep and systematic understanding of processes involved in conducting research in the area of advanced educational technologies. The course aims to equip the researchers with the fundamental concepts of database design methodologies required for the research areas in educational technologies.***

SECTION - A

Basics of Educational Technology: Concept and evolution of educational technology and information and communication technologies, components, architecture, models, characteristics of advanced learning and e-knowledge, dynamic content generation and management: services and technology.

Instruction, Communication, Media and Learning Designs: Changing trends to virtual/online interaction and Semantic Web; significance, critical perspective and use of design patterns for advanced learning; developing taxonomy for learning designs, learning design representations to document, model and share teaching practices.

Instructional Designs: General Guidelines for Instruction Design, Programmed instruction, Keller's plan, mastery learning, collaborative instruction, cooperative instruction, large group instructions, basic content research methodologies, Designing instructions for dissemination of information, the Dimensions of an Instructional Design Knowledge Base , General Systems Theory.

Standards for Advanced Educational Technology: SCROM, IMS etc; Principled construction and reuse of advance educational applications using standards, Design, Development and Evaluation of applications using techniques, Policies and priority areas of recent trends in Learning Systems.

SECTION - B

Introduction to Database Systems: Database System Applications, Normalization: 1NF, 2NF, 3NF, BCNF, 4NF and 5NF, SQL Queries, Database Users and Administrators, Transactions and its properties, Serializability: Conflict serializability, View serializability, Testing for serializability, Concurrency Control Techniques: Lock-Based Protocols, Timestamp Based Protocols, Validation based Protocol.

Recovery Systems and Oracle Concepts: Failure Classification, Storage Structure: Types, Stable Storage Implementation, Data Access, Log Based Recovery, Advanced recovery Techniques: Logical Undo Logging, Transaction Rollback, Checkpoints, Remote Backup Systems.

Introduction to SQL*PLUS, referential Integrity, SQL*PLUS Reports, Introduction to PL/SQL, Cursors, triggers, Procedures, Functions, packages, Large Objects, Creating Users, Remote Data Access.

Indexing and Concurrency Control: Types of Single-Level Ordered Indexes, Multilevel Indexes, Dynamic Multilevel Indexes Using B-Tress and B+Trees, Locking Techniques of Concurrency Control and Concurrency Control techniques Based on Timestamp Ordering

Distributed Databases- Distributed Data Storage, Network Transparency, Distributed Query Processing, Distributed transaction Model, Commit Protocols, Coordinator selection, Deadlock detection, recovery and handling

Suggestive Readings:

1. D'Antoni, S. & Savage, C. (2009). Open Educational Resources: Conversations in Cyberspace. New York: United Nations Educational, Scientific and Cultural Organization.
2. Ehlers, U.D. & Schneckenberg, D. (2010). Changing Cultures in Higher Education: Moving Ahead to Future Learning. London: Springer.
3. Jonassen, D.H. (ed) (2003). Learning to Solve Problems with Technology: A Constructivist Perspective, (Edition 2).
4. Merrill. Joyce, B.R., Weil, M. & Calhoun, E. (2009). Models of Teaching, Alternative eText Formats Series, (Edition 8). Boston: Pearson/Allyn and Bacon Publishers.
5. Kanvaria, V. K. (2014). A Comprehension on Educational Technology and ICT for Education.
6. Ledford, B.R. & Sleeman, P.J. (2001). Instructional Design: A Primer. Greenwich: Information Age Publishing. Leonard, D.C. (2002). Learning theories: A to Z. Westport: Greenwood Publishing Group.
7. Lori Lockyer,Sue Bennett Learning designs and Learning Objects Issues, Applications and Technologies Volume II , Information Science Press
8. Database System Concepts by A. Silberschatz, H.F Korth and S. Sudarshan,3rd ed,1997, McGraw-Hill and International Edition
9. Fundamentals of database Systems by R.Elmasri and S.B Navathe, 3rd edition
10. An introduction to Database Systems by C.J date, 7th ed., Addison-Wesley, Low Priced Edition,2000

Mobile Communication and Android Application Development

Paper Code: **CSA 108**

Objectives: The course will familiarize the researchers with concepts about mobile communication, its architecture, protocols, and mobile databases. It will also enable them to develop variety of mobile applications using Android.

SECTION–A

Mobile communication: Mobile computing architecture, Mobile Devices, Mobile System Networks: Cellular, WLAN, Ad hoc networks; GSM: Services and System Architecture, Radio Interfaces, Localization, Calling, Handover, Security; CDMA: Architecture, Spread Spectrum, Coding Methods; GPRS: Introduction, system Architecture; Introduction to EDGE, LTE.

Mobile database: Database hoarding techniques, Data Caching, Client Server computing: 2-tier and 3-tier client server architecture; Transactional models, Query processing, Data Recovery process; Classification of Data delivery mechanism: Push based, pull based, Hybrid; Selective tuning and indexing technique, comparative study of various Mobile Operating systems.

Android Application Development: Android architecture, Android versions and its feature set, Android Market application store, installation of tools and SDK for developing Android applications, Using Android Documentation, Working with Android Emulator.

SECTION–B

Building Android Applications: life cycle of an activity, creating User Interface (UI),working with activities, intents, dialogs, fragments etc., applying styles and themes to activities, displaying pictures and menus with views, managing application resources, configuring the Android Manifest file, registering activities, designating the launch activity, managing application permissions.

Android Data Persistence: saving simple data, writing and reading files in internal and external storage, Creating and using a SQLite database, building data base applications.

Android Networking: obtaining geographical data using GPS, Cell-ID, and Wi-Fi triangulation, monitoring for a location, building a Location Tracker application, connecting to web, using JSON web services: Publishing Android Applications.

Suggested Readings:

1. Kamal, Raj, Mobile Computing, Oxford Higher Education.
2. Jochen Schiller, “Mobile Communications”, PHI/Pearson Education, Second Edition.
3. Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober, “Principles of Mobile Computing”, Springer, New York, 2003.
4. Reto Meier, Professional Android 2 Application Development, Wrox Publications.
5. Wei-Meng Lee, Beginning Android Application Development,Wiley Publications.
6. Shane Conder, Lauren Darcey: Sams Teach Yourself Android™ Application Development in 24 Hours.

Parallel and Distributed Computing

Paper Code: **CSA 109**

Objectives: The course enables students to learn details of parallel and distributed computing. After completing the course, the candidate will be capable of performing research in parallel and distributed computing.

SECTION–A

Parallel Computing: Taxonomy of parallel processing. Parallel techniques and paradigms; Abstract models of parallel machines; Strategies and paradigms for parallel algorithm design; Correlations of algorithmic to architectural spaces;

Design methodologies for parallel algorithms for parallel computing platforms; Designing parallel algorithms for various application areas: linear algebra, matrix multiplication, sorting, PDE, graphs, combinatorial search, document classification;

Design methodologies for parallel genetic algorithms parallel programming with MPI, Parallel programming with OpenMP, Hybrid parallel programming MPI+OpenMP

Multithreaded programming, Parallel program benchmarking, Parallelism profiling

SECTION – B

Cluster Computing: The Role of Clusters, Architecture and Cluster Software, Network Hardware, Network Software, Protocols, Distributed File Systems, Virtualization technologies, Benchmarks

Grid Computing: Role of a Grid, infrastructure**,** Grid Architecture, Overview of Resource Managers, Application Management; Grid Application Description Languages, Application Partitioning, Meta-scheduling, Mapping ,Monitoring, Web Services, Grid Portals.

Cloud Computing: Fundamentals, private, Public and hybrid clouds, characteristics, examples, and applications, difference between cloud computing and distributed computing

Books:			
1.	Grama A., Gupta A., Karypis G., Kumar V	:	Introduction to Parallel Computing, PEARSON, Addison Wesley
2.	Barry Wilkinson, Michael Allen	:	Parallel Programming Techniques and Applications Using Networked Workstations and Parallel Computers Pearson Prentice Hall .
3.	Haupt R., S. Haupt	:	Practical Genetic Algorithms, Wiley Interscience, A John Wiley & Sons
4.	M. Quinn	:	Parallel Programming in C with MPI and OpenMP, McGraw Hill
5.	Hwang K., Zhiwei X.	:	Scalable Parallel Computing: Technology, Architecture, Programming, McGraw-Hill.
6.	Timothy Chou	:	Fundamentals of Cloud Computing, Cloudbook, Inc., 2015

Data Mining and Data Warehousing

Paper Code: **CSA 110**

Objectives : This course will introduce concepts and techniques of data mining and data warehousing, including concept, principle, architecture, design, implementation, application of data warehousing and data mining. Some systems for data warehousing and/or data mining will also be introduced to study the methodology of engineering legacy databases for data warehousing and data mining to derive business rules for decision support systems.

SECTION – A

Introduction : Data Warehousing, Characteristics of a Data Warehouse.

Developing Data Warehouse: Building a Data Warehouse, Data Warehouse architectural strategies, Design considerations, metadata, Tools for Data Warehousing, performance & Technological considerations, Crucial decisions in Designing a Data Warehouse.

Security : Security in Data Warehouses, Difficulties in implementation of Data Warehouses, Open issues in data Warehouses.

Data Marts: Data Marts, Types of Data Marts, Developing Data Mart based Data Warehouses.

SECTION – B

OLTP and OLAP Systems: Data Modeling, Star Schema for multidimensional view, Snowflake schema OLAP tools, Managed Query Environment. OLTP versus OLAP.

Data Mining : Introduction, Steps of Data Mining Process, Types of Data Mining Tasks, Data Mining Algorithms : Association (Aprior Algorithm) a nd Classification (Decision Tree Induction), Back propagation Neural Network, Clustering, Tools for Data Mining. Programming Using Standard & Open Source Data Warehousing and Data Mining Tools like ORACLE Data Mining, Hdoop and MATLAB.

Suggested Readings :

1. Cornel and Rao : "Database Principles" Cengage, Publications.
2. Benson Alex and Smith Stephen J. : Data Warehousing, Data Mining and OLAP, TMH.
3. Jiawei Han, Micheline Kamber, 2000 : Data Mining : Concepts and Techniques, Morgan Koffman Elscvier.
4. Inmon, W.H., 1999 : Managing the Data Warehouse, C. L. Gassey, John Wiley.
5. Pujari, Arun K. : Data Mining Techniques, University Press.
6. Mattison, 1999 : Data Warehousing and Knowledge Management, Tata McGraw Hill.
7. Fayyad, Usama M.,1996 : Advances in Knowledge Discovery and Data Mining, MIT Press.
8. ELMASI & NAVATHE : "Fundamentals of Database Systems" Pearson Education.

Simulation & Modeling

Paper Code: **CSA 111**

Objectives: The course aim to apprise the researchers with simulation and modeling techniques for effective research. After going through the course, the candidate will be able to choose and apply these techniques for their research areas.

SECTION – A

Introduction to Simulation: Simulation: Concepts, Areas of Application, Systems and System Environment, Components of a system, Discrete and Continuous Systems, Model of a System, Types of Models, Discrete-Event System Simulation, Steps in a Simulation Study.

System Studies: Subsystems, A Corporate Model, Environment Segment, Production Segment, Management Segment, The Full Corporate Model, Types of System Study, System Analysis, System Design, System Postulation.

System Simulation: The Technique of Simulation, The Monte Carlo Method, Comparison of Simulation and Analytical Methods, Experimental Nature of Simulation, Types of System Simulation, Numerical Computation Technique for Continuous Models, Distributed Lag Models, Cobweb Models.

System Dynamics: Exponential Growth Models, Exponential Decay Models, Modified Exponential Growth Models, Logistic Curves, System Dynamics Diagrams, Simple System Dynamics Diagrams, Multi-Segment Models, Representation of Time Delays.

SECTION – B

Probability Concepts in Simulation: Stochastic Variables, Discrete Probability Functions, Continuous Probability Functions, Measures of Probability Functions, Numerical Evaluation of Continuous Probability Functions, Continuous Uniformly Distributed Random Numbers, Computer Generation of Random Numbers, A Uniform Random Number Generator, Generating Discrete Distributions, Non-Uniform Continuously Distributed Random Numbers, The Rejection Method.

Arrival Patterns and Service Times: Congestion in Systems, Arrival Patterns, Poisson Arrival Patterns, The Exponential Distribution, The Coefficient of Variation, The Erlang Distribution, The Hyper-Exponential Distribution, Service Times, The Normal Distribution, Queuing Disciplines, Queuing notation, Measures of Queues, Mathematical Solutions of Queuing Problems.

Discrete System Simulation: Discrete Events, Representation of Time, Generation of Arrival Patterns, Case Study of System Simulations, Simulation Programming Tasks, Gathering Statistics, Counters and Summary Statistics, Measuring Utilization and Occupancy, Recording Distributions and Transit Times, Discrete Simulation Languages.

Input Modelling: Data Collection, Identifying the Distribution with Data, Parameter Estimation, Selecting Input Models without Data.

Visualization: Fundamentals, techniques and applications.

Finite Element Analysis: Finite Element method and practical application to different problems

Simulation Software: Simulation in C++,Other open source simulations tools like NS3

Suggested Readings :

1. Averil,M.Law, W. David Kelton- Simulation Modeling and Analysis, 3/e,Mc Graw Hill Publications 2000
2. C.S. Krishnamoorthy- Finite Element Analysis: Theory and Programming
3. Jerry Banks, "Handbook of Simulation: Principles, Methodology, Advances, Applications, and Practice", John Wiley & Sons, Inc., 1998.
4. George S. Fishman, "Discrete-Event Simulation: Modeling, Programming and Analysis", Springer-Verlag New York, Inc., 2001.
5. Andrew F. Seila, Vlatko Ceric, Pandu Tadikamalla, "Applied Simulation Modeling", Thomson Learning Inc., 2003.

Network Technologies

Paper Code: **CSA 112**

Objectives: This challenging and comprehensive course provides a broad perspective on the networks and related technologies.

SECTION–A

Overview of wireless networks: Introduction to wireless communication, Cellular Networks and generations, IEEE 802.11, IEEE 802.16 and IEEE 802.20.

IPv6: Basic protocol, extensions and options, support for Quality of Service (QoS), security, neighbor discovery, auto-configuration, routing.

Traffic Management: Economic Framework, Traffic Models, Traffic Classes, Scheduling, Admission Control, Peak Load Pricing.

QoS and Security issues: Network security requirements, issues and challenges in security and QoS provisioning, classifications of QoS solutions, IP security overview, IP Security Architecture.

SECTION – B

Mobility in networks: Mobile IP and related issues like Route Optimization, Handoff and Security.

Mobile Ad hoc Networks (MANETs): Introduction to Ad hoc wireless networks and sensor networks, Various MANET issues.

Transport layer: Various transport layer solutions and TCP over Ad hoc wireless networks, TCP extensions for high-speed networks, other new options in TCP, TCP in Wireless Domain.

Network Simulation: Elements of queuing model, role of exponential distribution, Event graphs of queuing model, Discrete-event and Continuous Simulation, Introduction to NS3 and Qualnet.

Books:

1. William Stallings : Wireless Communication and Networks, Pearson
2. C. Siva Ram Murthy and B. S Manoj : Adhoc Wireless Networks-Architecture and Protocols, Pearson Education
3. W. R. Stevens : TCP/IP Illustrated, Vol. 1, Pearson Education
4. M. Gonsalves and K. Niles : IPv6 Networks, McGraw Hill
5. S. Keshav : An Engineering Approach to Computer Networking, Pearson Education
6. Kamal, Raj Mobile Computing, Oxford Higher Education.
7. Jochen Schiller Mobile Communications, PHI/Pearson Education, Second Edition.
8. Requests for Comments (RFCs) & Internet Drafts, published by Internet Engineering Task Force (www.rfc-editor.org)
9. Journals: IEEE Journal on Selected Areas in Communications, IEEE Transactions on Communication, ACM/IEEE Transactions on Networking